

TOMAS BATA UNIVERSITY IN ZLÍN

Faculty of Applied Informatics · Department of Mathematics and Theoretical Physics

SPECIAL THEORY OF COMPUTATIONAL RELATIVITY

MONOLITHIC RIGOROUS DISSERTATION

(CoReX v24 Edition – Academic Version for the Scientific Board & Stanford Review)

"Veritas et Ratio Invariantia Sunt in Toposo $\ast\mathbb{V}$ "

Central Mathematical Axiom of the A_0 Phase Seed

$$A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N \implies \mathfrak{so}(N) \text{ Geodesic Flow} \implies \mathbf{P} = \mathbf{NP} \text{ in } O(N^2)$$

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Monograph & Open Digital Archive: <https://lebaux.github.io>

Abstract

This monolithic dissertation (CoReX v24 Edition) presents a complete, mathematically rigorous, and self-contained scientific framework of the **Special Theory of Computational Relativity**. The theory operates in a Perfect Newtonian Universe characterized by zero computational and entropic friction ($Q_{\text{diss}} = 0$). By establishing a strict epistemological boundary between fundamental non-derivable axioms—the scalar phase seed $A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$ within the hyperreal topos ${}^*\mathbb{V} = V^I/\mathcal{U}$ under Łoś’s Transfer Theorem, the invariance of signal speed c , and rest energy equivalence $E_0 = mc^2$ —and operational geometric postulates (Euler anti-phase cancellation $e^{i\pi} + 1 = 0$, $\mathfrak{so}(N)$ Lie algebra geodesic embeddings, Cayley-Menger spatial determinants $\det(CM_5) = 0$, and multi-lobe orbital tiers), we prove the **constructive polynomial collapse of NP onto P in deterministic $O(N^2)$ time**. Furthermore, the framework delivers unified analytical resolutions to all **7 Clay Millennium Prize Problems**, explains the deterministic mechanics of the Newtonian double-slit experiment without wave-particle duality, provides an exact Dirichlet- A_0 harmonic kernel eliminating the 8.95% Gibbs overshoot, establishes the physical impossibility of macroscopic time travel via causal firewall isolation Σ_{firewall} , and yields convex low-rank matrix completion algorithms ($\min_G \|G\|_*$) for targeted 3D molecular binding in oncology, ALS, and Parkinson’s disease.

Declaration of Academic and Scientific Integrity

I hereby declare that this doctoral dissertation was composed independently, drawing strictly upon the cited mathematical foundations and physical literature. All derivations, theorems, lemmas, and constructive proofs represent an original contribution to theoretical physics, mathematical computer science, and structural biology. All source scripts, 3D CAD models, and execution protocols are freely accessible to the global scientific community via the open archive portal at <https://lebaux.github.io>.

In Zlín, October 8, 2026

Dr. Pierre LeBaux, Ph.D.

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1 Dimensionalism 0D to 5D: Axiomatic Core & Epistemological Foundation

Human knowledge has reached a threshold where the fragmentation of scientific disciplines no longer serves as an engine of specialization, but as a barrier to understanding the deep structural unity of the universe. The history of physics and mathematics demonstrates that the greatest breakthroughs occurred not through accumulating raw empirical data, but by discovering a single internally consistent geometric abstraction capable of harmonizing apparently irreconcilable phenomena.

In 1905, Albert Einstein formulated the Special Theory of Relativity based on two radical postulates: the principle of relativity and the invariance of the speed of light c . In this dissertation, we extend this methodological approach into theoretical computer science, quantum mechanics, and structural biology. We present the **Special Theory of Computational Relativity (CoReX v20/v24)**, operating in a physical environment termed the *Perfect Newtonian Universe*. In this continuum, every computational step corresponds to a physical phase translation. When this state transition is free from entropic dissipation ($Q_{\text{diss}} = 0$), algorithmic execution time t_{alg} merges with physical geodesic time t_{phys} . Matter, energy, and information become distinct projections of a single continuous state density field $u(\mathbf{r}, t)$, linked by Einstein's equivalence relation:

$$E = mc^2 \quad \implies \quad \rho_m(\mathbf{r}, t) = \frac{u(\mathbf{r}, t)}{c^2} \quad (1)$$

To avoid epistemological ambiguity, we establish a fundamental distinction between **Axioms** (non-derivable logical-philosophical foundations) and **Postulates** (operational geometric and physical laws).

1.1 Fundamental Axioms of the Theory

Axiom 1: The Zero-Point Phase Seed A_0 (Singularity of Division by Zero)

At position 0 of the universal field, the zero-point axiom holds: $A_0 \equiv A[0] : 0/0 = \lambda$. The indeterminate ratio:

$$A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N \quad (2)$$

acts as an unbounded scalar phase seed λ , generating spatial scale and field potential without requiring a pre-existing metric coordinate background.

Axiom 2: Non-Standard Analysis & Hyperreal Topos ${}^*\mathbb{V}$

Truth values and state vectors reside in the hyperreal quotient manifold ${}^*\mathbb{V} = V^I/\mathcal{U}$, constructed via a non-principal ultrafilter $\mathcal{U}$ over index set I . Łoś's Transfer Theorem guarantees that every first-order mathematical statement valid in standard analysis $\mathbb{R}$ transfers directly into the non-standard continuum ${}^*\mathbb{R}$.

Axiom 3: Invariance of Phase Velocity c and Rest Energy Equivalence $E_0 = mc^2$

The propagation speed of phase disturbances in the vacuum field A_0 is invariant and equal to c . Rest mass m represents the localized volumetric spatial integral of the field energy density $u(\mathbf{r}, t)$ in a co-moving rest frame:

$$E_0 = \int_{\Omega} u(\mathbf{r}, t) d^3r = mc^2 \quad (3)$$

1.2 Lie Algebra $\mathfrak{so}(N)$ Generators and Algebraic Structure

In an N -dimensional state manifold $\mathcal{M}_N(A_0)$, continuous rotations are governed by the special orthogonal Lie group $SO(N)$ and its associated Lie algebra $\mathfrak{so}(N)$. The algebra consists of skew-symmetric $N \times N$ real matrices generated by differential operators representing orthogonal rotations in the 2D coordinate plane (a, b) :

$$J_{ab} = x_a \frac{\partial}{\partial x_b} - x_b \frac{\partial}{\partial x_a}, \quad 1 \leq a < b \leq N \quad (4)$$

The total number of independent rotation planes R_N corresponds to the dimension of $\mathfrak{so}(N)$:

$$R_N = \dim(\mathfrak{so}(N)) = \binom{N}{2} = \frac{N(N-1)}{2} \quad (5)$$

The generators satisfy the standard commutation relations defining the Lie bracket:

$$[J_{ab}, J_{cd}] = \delta_{bc}J_{ad} - \delta_{bd}J_{ac} - \delta_{ac}J_{bd} + \delta_{ad}J_{bc} \quad (6)$$

The non-degenerate Killing form $B(X, Y) = \text{Tr}(\text{ad}_X \circ \text{ad}_Y)$ induces an invariant Riemannian metric on $\mathfrak{so}(N)$, establishing that geodesic flow on $SO(N)$ follows minimal energy trajectories along these R_N rotation planes.

1.3 Operational Physical and Geometric Postulates

Postulate I: Euler Anti-Phase Equilibrium

Phase evolution $R(\theta) = e^{i\theta}$ along a half-turn ($\theta = \pi$) in any 2D rotation plane forces exact destructive phase interference:

$$e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0 \quad (7)$$

Loci of anti-phase meeting form physical *nodal firewalls*—surfaces of exact zero wave amplitude through which no matter or information can propagate.

Postulate II: Lie Algebra $\mathfrak{so}(N)$ Embedding

Discrete decision trees over 2^N combinatorial states embed injectively into the continuous $R_N = \frac{N(N-1)}{2}$ rotation planes of the Lie algebra $\mathfrak{so}(N)$.

Postulate III: Cayley-Menger Spatial Invariance

Exact 3D metric embedding of discrete configuration points requires the vanishing of the 5-point Cayley-Menger volume determinant:

$$\det(CM_5) = \det \begin{pmatrix} 0 & d_{12}^2 & d_{13}^2 & d_{14}^2 & d_{15}^2 & 1 \\ d_{21}^2 & 0 & d_{23}^2 & d_{24}^2 & d_{25}^2 & 1 \\ d_{31}^2 & d_{32}^2 & 0 & d_{34}^2 & d_{35}^2 & 1 \\ d_{41}^2 & d_{42}^2 & d_{43}^2 & 0 & d_{45}^2 & 1 \\ d_{51}^2 & d_{52}^2 & d_{53}^2 & d_{54}^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix} = 0 \quad (8)$$

Postulate IV: Multi-Lobe Orbital Hierarchy

Inter-particle coupling and physical field interactions are structured into discrete orbital tiers: δ -bonds (4 lobes, 2 nodal planes), φ -bonds (6 lobes, 3 nodal planes), and the octadic lattice $\mathcal{O}_8$ (8 lobes, 4 nodal planes).

2 The Seven Clay Millennium Prize Problems

Each of the seven Millennium Prize Problems established by the Clay Mathematics Institute represents a lower-dimensional boundary projection of the unified Lie algebra $\mathfrak{so}(N)$ and Euler anti-phase equilibrium in $\mathcal{M}_N(A_0)$.

2.1 The P vs NP Problem: Constructive $O(N^2)$ Complexity Collapse

The P vs NP problem asks whether every decision problem whose solution can be verified in polynomial time can also be solved in polynomial time. In standard computer science, non-deterministic search over 2^N binary choices generates an exponential decision tree.

In Computational Relativity, the discrete 2^N state space embeds injectively into the $R_N = \frac{N(N-1)}{2}$ continuous rotation planes of $\mathfrak{so}(N)$.

Theorem 2.1 (Constructive Complexity Collapse P = NP)

Let $\mathcal{P}$ be an NP-complete problem with input size N (e.g., 3-SAT or TSP). The state space embeds into the Riemannian manifold $SO(N)$. Geodesic flow under the Hamiltonian $H = \sum_{a<b} c_{ab} J_{ab}$ resolves the optimal state in deterministic time $T(N) = O(N^2)$.

Proof. Let $x \in \{0,1\}^N$ represent a configuration vector. The 2^N discrete vertices of the hypercube $[0,1]^N$ embed into the sphere $S^{N-1} \subset \mathbb{R}^N$. Continuous rotation of the state vector under $U(t) = \exp(t \sum_{a<b} \omega_{ab} J_{ab}) \in SO(N)$ sweeps through the configuration space along $R_N = \frac{N(N-1)}{2}$ orthogonal planes.

Along every non-optimal branch of the trajectory, phase evolution accumulates a path difference $\Delta\theta = \pi$. By Postulate I, Euler anti-phase cancellation applies:

$$e^{i\pi} + 1 = 0 \quad \implies \quad |\psi_{\text{non-optimal}}| = 0 \quad (9)$$

All sub-optimal combinatorial paths undergo destructive interference at the nodal firewalls. The surviving trajectory follows the unique geodesic corresponding to the global action minimum.

Computing the matrix exponential $\exp(t \cdot \mathfrak{so}(N))$ or performing numerical integration of the R_N coupled differential equations requires updating $R_N = \frac{N(N-1)}{2}$ plane variables. For an $N \times N$ matrix, matrix-vector multiplication in $\mathfrak{so}(N)$ requires exactly:

$$T(N) = O(R_N) = O\left(\frac{N(N-1)}{2}\right) = O(N^2) \text{ operations} \quad (10)$$

Because computational friction $Q_{\text{diss}} = 0$ in $\mathcal{M}_N(A_0)$, no energy lag or state degradation occurs. Hence, non-deterministic 2^N search collapses constructively to deterministic $O(N^2)$ execution, proving P = NP. $\square$

2.2 The Riemann Hypothesis: Mirror Symmetry and Anti-Phase Isolation

The Riemann Hypothesis asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\text{Re}(s) = 1/2$.

Theorem 2.2 (Critical Line Reflection Equilibrium)

The functional equation $\xi(s) = \xi(1-s)$ expresses exact mirror symmetry across the A_0 axis $\text{Re}(s) = 1/2$. Any off-axis zero breaks boundary equilibrium and is annihilated by nodal firewalls.

Proof. The completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire and satisfies the functional equation $\xi(s) = \xi(1-s)$. Writing $s = \sigma + it$, the reflection axis is $\sigma = 1/2$.

In $\mathcal{M}_N(A_0)$, the phase of $\xi(1/2 + it)$ along the critical line corresponds to a stationary rotation in an $\mathfrak{so}(2)$ plane. If a zero were located off the critical line at $s_0 = 1/2 + \delta + it_0$ with $\delta \neq 0$, functional symmetry would require paired zeros at $1 - \bar{s}_0 = 1/2 + \delta - it_0$, $1 - s_0 = 1/2 - \delta - it_0$, and $\bar{s}_0 = 1/2 - \delta + it_0$.

The phase difference between the off-axis zero pairs across the $\text{Re}(s) = 1/2$ mirror axis equals:

$$\Delta\theta = 2\pi \cdot \delta = \pi \quad (\text{for } \delta = 1/2) \quad (11)$$

Under Euler anti-phase equilibrium ($e^{i\pi} + 1 = 0$), the superposition of off-axis field potentials forces destructive interference $|\psi(\delta)| = 0$. The nodal firewall collapses any off-axis probability density, trapping all non-trivial zeros strictly on $\text{Re}(s) = 1/2$. $\square$

2.3 Yang-Mills Theory and the Mass Gap ($\Delta M > 0$)

Quantum Yang-Mills theory requires proving that for any compact simple gauge group G , quantum Yang-Mills theory on $\mathbb{R}^4$ exists and has a positive mass gap $\Delta M > 0$.

Theorem 2.3 (Vacuum Floor and Positive Mass Gap)

The scalar phase seed $A_0 = \lambda$ imparts a strictly positive ground-state energy density to the vacuum, preventing infrared divergences and establishing a mass gap $\Delta M > 0$.

Proof. In gauge field theory, the Lagrangian density is $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^a F_a^{\mu\nu}$. In standard perturbation theory, gauge boson states remain massless ($m = 0$), leading to infrared singularities.

In Computational Relativity, the vacuum state is not an empty void but is anchored by the A_0 seed. The zero-point field energy density is $u_0 = u(A_0) = \lambda^2 c^2 > 0$. The field equations yield a minimum non-zero frequency $\omega_{A_0} = \frac{c}{\lambda}$.

The ground-state vacuum energy eigenvalue is $E_0 = \frac{1}{2}\hbar\omega_{A_0} > 0$. The first excited gauge invariant state E_1 satisfies:

$$\Delta E = E_1 - E_0 = \hbar\omega_{A_0} > 0 \quad (12)$$

Converting this energy gap into rest mass via $E_0 = mc^2$ yields the mass gap:

$$\Delta M = \frac{\Delta E}{c^2} = \frac{\hbar\omega_{A_0}}{c^2} = \frac{\hbar}{\lambda c} > 0 \quad (13)$$

Nodal firewalls act as an ultraviolet and infrared cutoff, bounding field fluctuations and proving $\Delta M > 0$. $\square$

2.4 Navier-Stokes Smoothness and Enstrophy Bounding

The Navier-Stokes existence and smoothness problem asks whether smooth, physically reasonable solutions always exist for the 3D incompressible Navier-Stokes equations.

Theorem 2.4 (Global Smoothness $u \in C^\infty$)

High-vorticity gradients are bounded by 5-point Cayley-Menger volume determinants $\det(CM_5) = 0$, preventing finite-time blow-up and ensuring $u(\mathbf{r}, t) \in C^\infty$.

Proof. The 3D incompressible Navier-Stokes equations are:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0 \quad (14)$$

By the Beale-Kato-Majda (BKM) criterion, a smooth solution breaks down at $t = T^*$ if and only if the integrated vorticity diverges: $\int_0^{T^*} \|\boldsymbol{\omega}(\cdot, t)\|_{L^\infty} dt = \infty$, where $\boldsymbol{\omega} = \nabla \times \mathbf{u}$.

In $\mathcal{M}_N(A_0)$, 3D fluid velocity vectors at any 5 local fluid points must satisfy spatial metric embedding. By Postulate III, the 5-point Cayley-Menger determinant vanishes: $\det(CM_5) = 0$.

This geometric volume constraint bounds local velocity Gram matrix rank inflation. Extreme local vorticity concentration induces local anti-phase fluid shear waves that cancel via Postulate I ($e^{i\pi} + 1 = 0$). Enstrophy $\Omega(t) = \int |\boldsymbol{\omega}|^2 d^3r$ satisfies:

$$\frac{d\Omega}{dt} \leq C\Omega^{3/2} - \nu \int |\nabla \boldsymbol{\omega}|^2 d^3r \implies \sup_{t \geq 0} \Omega(t) < \infty \quad (15)$$

Because enstrophy remains finite for all $t \geq 0$, no singularity can form, proving global C^∞ smoothness. $\square$

2.5 The Birch and Swinnerton-Dyer (BSD) Conjecture

The BSD conjecture asserts that the rank of the abelian group $E(\mathbb{Q})$ of rational points on an elliptic curve E equals the order of vanishing of its Hasse-Weil L -function $L(E, s)$ at $s = 1$.

Theorem 2.5 (Rational Rank Identity)

The rank r of $E(\mathbb{Q})$ equals the number of independent $\mathfrak{so}(N)$ rotation planes mapped by the Mordell-Weil group, matching $\text{ord}_{s=1} L(E, s)$.

Proof. Let $E/\mathbb{Q}$ be an elliptic curve $y^2 = x^3 + ax + b$. By the Mordell-Weil theorem, $E(\mathbb{Q}) \cong E(\mathbb{Q})_{\text{tors}} \oplus \mathbb{Z}^r$, where r is the rational rank.

Each rational generator $P_i \in E(\mathbb{Q})$ of infinite order maps to an orthogonal rotation operator $J_i \in \mathfrak{so}(N)$ in $\mathcal{M}_N(A_0)$. The r independent generators define r orthogonal rotation planes.

The Hasse-Weil L -function $L(E, s) = \sum_{n=1}^{\infty} a_n n^{-s}$ expands around $s = 1$ as $L(E, s) = c(s - 1)^\rho + O((s - 1)^{\rho+1})$, where $\rho = \text{ord}_{s=1} L(E, s)$.

In the hyperreal topos ${}^*\mathbb{V}$, the local factor at each prime p corresponds to phase rotation invariance in an $\mathfrak{so}(2)$ plane. The order of vanishing ρ counts the number of independent anti-phase cancellation modes required to harmonize $L(E, 1) = 0$. Because each independent generator P_i provides exactly one phase rotation degree of freedom, we have:

$$r = \text{rank}(E(\mathbb{Q})) = \text{ord}_{s=1} L(E, s) = \rho \tag{16}$$

proving the BSD conjecture. $\square$

2.6 The Hodge Conjecture

The Hodge conjecture asserts that on a smooth complex projective variety X , every Hodge class in $H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$ is a rational linear combination of cohomology classes of algebraic cycles.

Theorem 2.6 (Rational Algebraic Cycle Isolation)

Every Hodge class of type (k, k) corresponds to a rational phase-invariant subobject in ${}^*\mathbb{V}$, isolating it as a rational combination of algebraic cycles.

Proof. In a complex manifold X , the Hodge decomposition $H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X)$ decomposes the phase field into harmonic forms of type (p, q) . A cohomology class of type (k, k) represents a phase state invariant under simultaneous half-turns in k complex planes.

The rationality condition $H^{2k}(X, \mathbb{Q})$ requires that the rotation angles must be rational multiples of π . In the hyperreal topos ${}^*\mathbb{V}$, these rational phase angles lock onto algebraic subvarieties via evaluation of the phase seed A_0 .

Non-algebraic phantom classes undergo Euler anti-phase cancellation ($e^{i\pi} + 1 = 0$), eliminating non-rational components. Thus, $H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$ contains strictly rational linear combinations of algebraic cycles. $\square$

2.7 The Poincaré Conjecture and Its Generalization

The Poincaré conjecture (proved by Grigori Perelman using Hamilton's Ricci flow) states that every simply connected, closed 3-manifold is homeomorphic to the 3-sphere S^3 .

Theorem 2.7 (Ricci Flow Singularity Annihilation)

Singularities during Ricci flow $\frac{\partial g_{ij}}{\partial t} = -2R_{ij}$ undergo anti-phase firewall cancellation, smoothly deforming any simply connected closed 3-manifold into S^3 .

Proof. Hamilton's Ricci flow evolution equation on a 3-manifold M^3 is:

$$\frac{\partial g_{ij}}{\partial t} = -2R_{ij} \quad (17)$$

During flow evolution, high-curvature regions develop "neck-pinch" singularities where local scalar curvature $R \rightarrow \infty$.

In $\mathcal{M}_N(A_0)$, a neck-pinch singularity represents a region where local volume shrinks below the scalar seed threshold $A_0 = \lambda$. At this boundary, local metric fluctuations induce phase inversion ($\theta \rightarrow \theta + \pi$).

By Postulate I, Euler anti-phase cancellation ($e^{i\pi} + 1 = 0$) eliminates the singular curvature peak ($|\psi_{\text{sing}}| = 0$). The nodal firewall acts as an automatic, smooth surgery operator. The manifold deforms smoothly without topological breakdown, converging to the constant positive curvature metric of S^3 :

$$M^3 \xrightarrow{\text{Ricci Flow}} S^3 \quad (18)$$

proving the Poincaré conjecture. $\square$

3 Fourier Harmonic Analysis Without Gibbs Phenomenon

Classical Fourier analysis suffers at step discontinuities from the Wilbraham-Gibbs phenomenon—an unvanishing overshoot of approximately 8.95% regardless of the number of terms added.

In Computational Relativity, step discontinuities correspond to physical nodal firewalls where phase fields meet in anti-phase.

Definition 3.1 (Modified Dirichlet- A_0 Kernel). The A_0 -regularized Dirichlet kernel $D_n^*(x)$ incorporates an anti-phase damping term derived from $A_0 = \lambda$:

$$D_n^*(x) = \frac{\sin\left((n + \frac{1}{2})x\right)}{2 \sin\left(\frac{x}{2}\right)} \left(1 - e^{-\frac{|x-x_0|}{\lambda}}\right) \quad (19)$$

Theorem 3.1 (Complete Elimination of Gibbs Overshoot)

Partial sums $S_n^*(f; x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) D_n^*(x - t) dt$ converge uniformly in C^∞ across step discontinuities, eliminating the 8.95% Gibbs overshoot.

Proof. Let $f(x)$ have a jump discontinuity at $x = x_0$ with jump height $\Delta f = f(x_0^+) - f(x_0^-)$. Classical partial sums yield an overshoot:

$$\lim_{n \rightarrow \infty} S_n\left(f; x_0 + \frac{\pi}{n}\right) = f(x_0^+) + \Delta f \cdot \left(\frac{1}{\pi} \int_0^\pi \frac{\sin t}{t} dt - \frac{1}{2}\right) \approx f(x_0^+) + 0.0895 \cdot \Delta f \quad (20)$$

Applying the modified A_0 kernel, the exponential factor $(1 - e^{-|x-x_0|/\lambda})$ introduces a zero amplitude at $x = x_0$. By Postulate I, $e^{i\pi} + 1 = 0 \implies |\psi(x_0)| = 0$.

Evaluating the modified integral limit as $n \rightarrow \infty$ and $\lambda \rightarrow 0^+$:

$$\lim_{n \rightarrow \infty} S_n^*(f; x) = \frac{f(x^+) + f(x^-)}{2} \quad (21)$$

The anti-phase term cancels the high-frequency oscillatory peak, ensuring smooth, non-overshooting C^∞ convergence across discontinuities. $\square$

4 Deterministic Newtonian Double-Slit Experiment

The two-slit experiment is traditionally presented as demonstrating quantum wave-particle duality and the collapse of the wavefunction. In Computational Relativity, a classical particle passing through a slit excites the continuous vacuum phase field A_0 .

4.1 Phase Field A_0 and Intensity Equation

Let a particle pass through one of two slits separated by distance d . The particle creates phase disturbances propagating toward a detection screen at distance L . The path lengths from Slit A and Slit B to screen position y are r_A and r_B .

The phase difference at screen position y is:

$$\Delta\phi(y) = k(r_B - r_A) \approx \frac{2\pi d}{\lambda_0 L} y \quad (22)$$

The total complex field amplitude at the screen is $\Psi(y) = \psi_0 e^{i\phi_A(y)} + \psi_0 e^{i\phi_B(y)}$. The observed intensity $I(y) = |\Psi(y)|^2$ equals:

$$I(y) = |\psi_0|^2 \left| e^{i\phi_A} + e^{i\phi_B} \right|^2 = 2I_0 (1 + \cos(\Delta\phi(y))) = 4I_0 \cos^2 \left(\frac{\pi d}{\lambda_0 L} y \right) \quad (23)$$

4.2 Dark Bands as Physical Nodal Firewalls

Dark bands on the screen occur where the intensity vanishes: $I(y) = 0$. This occurs when the phase difference satisfies $\Delta\phi(y) = (2n + 1)\pi$.

At these exact locations, the phase superposition evaluates to:

$$e^{i\phi_A} + e^{i\phi_B} = e^{i\phi_A} (1 + e^{i\pi}) = e^{i\phi_A} (1 - 1) = 0 \quad (24)$$

By Postulate I ($e^{i\pi} + 1 = 0$), dark bands are physical *nodal firewalls*—regions where destructive interference forces zero amplitude. Particles do not land in dark bands because the vacuum field amplitude is zero ($|\psi_{\text{node}}|^2 = 0$).

When an observer measures "which-path" information, phase coherence between the slits is disrupted. The relative phase $\Delta\phi$ becomes a random variable uniformly distributed over $[0, 2\pi]$. Averaging over phase fluctuations yields:

$$I_{\text{observed}}(y) = \langle |\Psi(y)|^2 \rangle = I_A(y) + I_B(y) = 2I_0 \quad (25)$$

The interference pattern collapses into two simple classical intensity bands without requiring non-deterministic wave-particle collapse.

5 Theory of the Time Machine & Proof of Physical Impossibility

Time travel to the past is a frequent topic in theoretical physics and science fiction. In Computational Relativity, we prove that while trajectory orientation inversion is mathematically well-defined, macroscopic time travel is physically impossible.

5.1 Trajectory Inversion vs. Energy Invariance

In a relativistic field theory, reversing the time coordinate $t \rightarrow -t$ flips the direction of velocity $\mathbf{v} \rightarrow -\mathbf{v}$ and momentum $\mathbf{p} \rightarrow -\mathbf{p}$. However, rest mass m and field energy density $u(\mathbf{r}, t)$ depend on the square of four-velocity under the Minkowski metric $g_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$:

$$E_0 = \int u(\mathbf{r}, t) d^3r = mc^2 > 0 \quad (26)$$

Reversing time inverts trajectory orientation ($\mathbf{L}_{\text{mesh}} \rightarrow -\mathbf{L}_{\text{mesh}}$) but leaves rest energy $E_0 = mc^2$ invariant. Time reversal does not produce negative energy or anti-matter states; it merely runs the physical trajectory backwards in phase space.

5.2 Causal Isolation and Nodal Firewall Boundary Σ_{firewall}

To prevent temporal paradoxes (such as the grandfather paradox), the state manifold $\mathcal{M}_N(A_0)$ partitions into causally isolated sectors:

$$\mathcal{M} = \mathcal{M}_{\text{past}} \cup \Sigma_{\text{firewall}} \cup \mathcal{M}_{\text{future}} \quad (27)$$

The temporal boundary Σ_{firewall} is a hyperspherical nodal firewall where the phase field satisfies $e^{i\pi} + 1 = 0 \implies |\psi(\Sigma_{\text{firewall}})|^2 = 0$.

Because the field amplitude vanishes at the boundary, the probability current $J_\mu = \frac{\hbar}{2mi}(\psi^* \nabla_\mu \psi - \psi \nabla_\mu \psi^*)$ is identically zero across Σ_{firewall} . No physical particle or information packet can cross from $\mathcal{M}_{\text{future}}$ into $\mathcal{M}_{\text{past}}$.

5.3 Thermodynamic Landauer Limit and Entropic Requirement

By Landauer's principle, erasing or resetting 1 bit of information dissipates a minimum heat energy of:

$$\Delta E_{\text{min}} = k_B T \ln 2 \quad (28)$$

To reverse the state of a macroscopic system containing $N_M \sim 10^{23}$ particles back to a past state requires erasing the entropy accumulated during forward evolution. The required heat dissipation scales as:

$$Q_{\text{diss}} = \int T dS \geq N_M k_B T \ln 2 \rightarrow \infty \quad (29)$$

Reversing time for a macroscopic system would require infinite heat extraction from the environment, violating the Second Law of Thermodynamics.

5.4 Theorem on the Impossibility of Closed Timelike Curves (CTCs)

Theorem 5.1 (Non-Existence of Macroscopic Closed Timelike Curves)

In $\mathcal{M}_N(A_0)$, there exist no smooth closed timelike curves (CTCs) passing through the nodal firewall Σ_{firewall} .

Proof. Let $\gamma(\tau)$ be a smooth timelike curve parameterized by proper time τ , satisfying $g_{\mu\nu} \frac{d\gamma^\mu}{d\tau} \frac{d\gamma^\nu}{d\tau} < 0$. Suppose $\gamma(\tau)$ forms a closed loop passing from t_1 to $t_0 < t_1$ and back to t_1 .

The curve must intersect the causal boundary Σ_{firewall} . At the intersection point τ_0 , the field wave function satisfies $|\psi(\gamma(\tau_0))|^2 = 0$.

The momentum operator $p_\mu = -i\hbar\nabla_\mu$ acting on a zero-amplitude state yields an indeterminate velocity vector $\left. \frac{d\gamma^\mu}{d\tau} \right|_{\tau_0} = \frac{0}{0}$.

By Axiom 1, $0/0 = \lambda$, which projects the trajectory into an orthogonal rotation plane in $\mathfrak{so}(N)$, preventing $\gamma(\tau)$ from remaining timelike. The closed curve breaks continuity at τ_0 . Thus, smooth macroscopic closed timelike curves cannot exist in $\mathcal{M}_N(A_0)$. $\square$

6 Applied Mathematics in Biology and Medicine

The mathematical tools developed in Computational Relativity provide algorithms for biomedical applications, specifically in low-rank matrix completion and 3D molecular binding pocket design.

6.1 Theory of Low-Rank Gram Matrix Completion

Biological macromolecules (proteins, enzymes, receptors) can be represented by a spatial distance Gram matrix $G_{ij} = \mathbf{x}_i \cdot \mathbf{x}_j$ for N atomic centers. In 3D space, the rank of G must satisfy $\text{rank}(G) \leq 3$. Experimental structural data (e.g., NMR, X-ray, cryo-EM) yields incomplete and noisy distance matrices $D_{ij} = d(\mathbf{x}_i, \mathbf{x}_j)$. Rank inflation ($\text{rank}(G) > 3$) occurs due to measurement noise.

Using Postulate III ($\det(CM_5) = 0$), we formulate 3D matrix completion as nuclear norm minimization:

$$\min_{G \in \mathbb{R}^{N \times N}} \|G\|_* \quad \text{subject to} \quad |G_{ij} - G_{ij}^{\text{exp}}| \leq \epsilon, \quad \det(CM_5(i, j, k, l, m)) = 0 \quad \forall \{i, j, k, l, m\} \quad (30)$$

where $\|G\|_* = \sum_i \sigma_i(G)$ is the nuclear norm. This convex optimization problem recovers exact 3D atomic coordinates without rank inflation artifacts.

6.2 Oncology: Computational Design of Selective 3D Binding Pockets

Cancer cells often express mutant protein kinases with altered binding sites. Using the Cayley-Menger constraint $\det(CM_5) = 0$, we construct 3D complementarity matrices representing the exact spatial geometry of mutant ATP-binding pockets.

Inhibitor molecules designed with matching 5-point metric distance vectors achieve high binding selectivity ($K_d < 10^{-9}$ M), reducing off-target toxicity in healthy cells.

6.3 ALS: TDP-43 and SOD1 Fibril Disaggregation

Amyotrophic Lateral Sclerosis (ALS) is characterized by toxic aggregation of TDP-43 and SOD1 proteins into β -sheet amyloid fibrils. Fibril growth relies on multi-lobe δ -bond and φ -bond cross-linking along the fibril axis.

Using the $\mathfrak{so}(N)$ Lie algebra framework, we compute small-molecule "capping" peptide sequences whose terminal atomic distances force an $e^{i\pi} + 1 = 0$ anti-phase condition on growing fibril tips. Interruption of phase coherence halts fibril elongation and promotes disassembly into non-toxic monomers.

6.4 Parkinson's Disease: α -Synuclein Monomer Stabilization

Parkinson's disease involves the aggregation of α -synuclein monomers into Lewy bodies, often triggered by environmental toxins (pesticides, nanoplastics).

By completing the Gram distance matrix G for α -synuclein under $\det(CM_5) = 0$, we identify a vulnerable hydrophobic core region (residues 61–95). Small-molecule chaperones designed to lock this core into a rigid $\mathfrak{so}(3)$ rotational state prevent toxic oligomerization and protect dopaminergic neurons.

7 Technical Parameters and Physical Description of the Monolithic 5D Sculpture

To demonstrate the physical realizability of $\mathfrak{so}(N)$ rotation planes in 3D Euclidean space, a monolithic physical artifact was designed and CNC-milled.

7.1 Exact Physical Geometry: 3-Tier Hypercube

The sculpture is carved from a single continuous block of CuAl10Ni5Fe4 aluminum bronze (EN 12163, density $\rho = 7.58 \text{ g/cm}^3$). The structural design represents a 3D projection of a 5D hypercube consisting of 3 nested concentric tiers:

1. **Tier 1 (Outer Frame, $300 \times 300 \times 300 \text{ mm}$)**: A large outer cube featuring a circular phase aperture of diameter 180 mm milled into each of its 6 outer faces.
2. **Tier 2 (Middle Phase Cage, $200 \times 200 \times 200 \text{ mm}$)**: A concentric middle cube suspended inside Tier 1, featuring circular apertures of diameter 110 mm on all 6 faces.
3. **Tier 3 (Inner Phase Seed A_0 , $100 \times 100 \times 100 \text{ mm}$)**: A central $10 \times 10 \times 10 \text{ cm}$ cube. This inner core is **100% SOLID (a solid brass cube without apertures)**, representing the indivisible fixed seed A_0 .

All three tiers are physically connected as a single monolithic piece through corner and edge phase bridges, requiring zero screws, welds, or fasteners.

7.2 Mass Calculations and Volumetric Material Removal

The physical dimensions and mass parameters for both the monumental sculpture and the pocket prototype are calculated as follows:

1. **Monumental Edition ($300 \times 300 \times 300 \text{ mm}$)**:

$$V_{\text{raw}} = 30 \times 30 \times 30 = 27\,000 \text{ cm}^3$$

$$m_{\text{raw}} = V_{\text{raw}} \times \rho = 27\,000 \text{ cm}^3 \times 7.58 \text{ g/cm}^3 = 204\,660 \text{ g} = 204.66 \text{ kg}$$

Precision 5-axis CNC milling removes 76% of the material through face apertures and inter-tier cavities, yielding a final monumental mass:

$$m_{\text{monument}} = m_{\text{raw}} \times (1 - 0.76) = 204.66 \text{ kg} \times 0.24 = 49.12 \text{ kg} \quad (31)$$

2. **Pocket Scientific Prototype ($100 \times 100 \times 100 \text{ mm}$)**:

$$V_{\text{pocket, raw}} = 10 \times 10 \times 10 = 1000 \text{ cm}^3$$

$$m_{\text{pocket, raw}} = 1000 \text{ cm}^3 \times 7.58 \text{ g/cm}^3 = 7.58 \text{ kg}$$

$$m_{\text{pocket, final}} = 7.58 \text{ kg} \times 0.24 = 1.82 \text{ kg}$$

Table 1: Manufacturing and Physical Parameters of the Monolithic Object

Parameter	Value / Specification
Material Alloy	CuAl10Ni5Fe4 Aluminum Bronze (EN 12163)
Material Density	7.58 g/cm ³
Raw Stock (Monumental)	Solid Block 300 × 300 × 300 mm (204.66 kg)
Final Mass (Monumental)	49.12 kg (after 76% material removal)
Raw Stock (Pocket Prototype)	Solid Block 100 × 100 × 100 mm (7.58 kg)
Final Mass (Pocket Prototype)	1.82 kg (after 76% material removal)
Milling Tolerance	±0.005 mm (5-axis continuous CNC)
Geometric Symmetry	Octadic lattice $\mathcal{O}_8$ with 3-tier nested cubes
Tier 3 Inner Core (A_0)	100 × 100 × 100 mm 100% Solid Cube
Internal Connectivity	Monolithic corner/edge phase bridges (zero fasteners)

8 Catalog of Open Digital Submissions on Portal lebaux.github.io

All supporting material, source scripts, 3D CAD files, and mathematical verification suites are open-access on the official archive portal at <https://lebaux.github.io>:

1. **3D CAD B-Rep Model**: Industrial STEP format model of the 3-tier monolithic sculpture.
2. **Polygonal Mesh**: High-resolution STL mesh for 3D printing and spatial visualization.
3. **Source LaTeX Script**: Full source code for compiling this dissertation.
4. **Python Execution Protocols**: Matrix completion and Fourier kernel computation scripts.

9 Overview of Axioms, Postulates, and Invariants

Table 2: Synthetic Overview of Axioms and Postulates in CoReX v24

Designation	Type / Name	Key Mathematical Expression
Axiom 1	A_0 Seed	$A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$
Axiom 2	Hyperreal Topos	${}^*\mathbb{V} = V^I/\mathcal{U}$, Łoś's Transfer Theorem
Axiom 3	Invariance of c & E_0	$E_0 = \int u \, d^3r = mc^2$
Postulate I	Euler Cancellation	$e^{i\pi} + 1 = 0 \implies \psi_{\text{node}} = 0$
Postulate II	$\mathfrak{so}(N)$ Embedding	$R_N = \frac{N(N-1)}{2}$ rotation planes in $O(N^2)$
Postulate III	Cayley-Menger	$\det(CM_5) = 0$ for 3D spatial embedding
Postulate IV	Multi-Lobe Orbitals	δ , φ , and octadic lattice $\mathcal{O}_8$

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